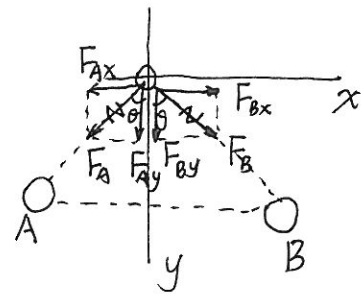
Phys. 2110 HW12 : 12.13, 39, 40, 41, 49, 55, 61, 70, 72, 89

12.13

$$F_A = G\frac{m_A m}{r^2} = G\frac{(0.26\,kg)(0.010\,kg)}{(0.100\,m)^2} = 1.735\times10^{-11}\,N$$

$$F_B = G\frac{m_B m}{r^2} = 1.735\times10^{-11}\,N$$



$$F_{Ax} = -F_A\sin\theta = -(1.735\times10^{-11}N)(0.80) = -1.39\times10^{-11}\,N$$

$$F_{Ay} = +F_A\cos\theta = +(1.735\times10^{-11}N)(0.60) = +1.04\times10^{-11}\,N$$

$$F_{Bx} = +F_B\sin\theta = +1.39\times10^{-11}N$$

$$F_{By} = +F_B\cos\theta = +1.04\times10^{-11}N$$

$\Sigma F_x = ma_x$ gives $F_{Ax} + F_{Bx} = ma_x$ $\quad a_x = 0$

$\Sigma F_y = ma_y$ gives $F_{Ay} + F_{By} = ma_y$

$$2(1.04\times10^{-11}N) = (0.010\,kg)\,a_y$$

$a_y = 2.1\times10^{-9}\,m/s^2$, directed downward midway between A and B

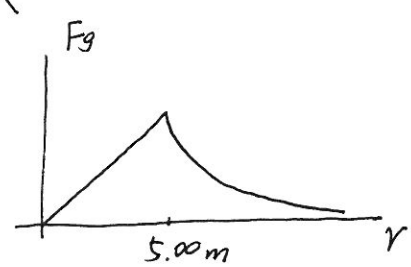
12.39 (a) (i) $F_g = \frac{GMm}{r^2} = (6.67\times10^{-11}N\cdot m^2/kg^2)\frac{(1000.0\,kg)(2.00\,kg)}{(5.01\,m)^2} = 5.31\times10^{-9}N$

(ii) $F_g = \frac{GM'm}{r^2}$. $\quad M' = M\left(\frac{r}{R}\right)^3 = (1000.0\,kg)\left(\frac{2.50m}{5.00m}\right)^3 = 125\,kg$

$$F_g = (6.67\times10^{-11}N\cdot m^2/kg^2)\frac{(125\,kg)(2.00\,kg)}{(2.50m)^2} = 2.67\times10^{-9}N$$

(b) $F_g = \frac{GM(r/R)^3 m}{r^2} = \left(\frac{GMm}{R^3}\right)r$ for $r<R$

$F_g = \frac{GMm}{r^2}$ for $r>R$



12.40 (a) $dU = -\frac{Gm\,dM}{L+x} = -\frac{GmM}{L}\frac{dl}{l+x}$

Integrating, $U = -\frac{GmM}{L}\int_0^L \frac{dl}{l+x} = -\frac{GmM}{L}\ln\left(1+\frac{L}{x}\right)$

For $x \gg L$, $\ln\left(1+\frac{L}{x}\right) \sim \frac{L}{x}$, and $U \to -\frac{GmM}{x}$

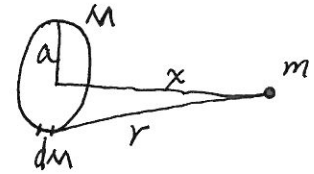
(b) $F_x = -\frac{\partial U}{\partial x} = \frac{GmM}{L}\frac{(-L/x^2)}{(1+L/x)} = -\frac{GmM}{x^2+Lx}$

The minus sign indicates an attractive force.

As $x \gg L$, $F_x \to -\frac{GmM}{x^2}$

12.41 (a) $dU = -\frac{Gm\,dM}{r}$

$$U = \int dU = -Gm\int \frac{dM}{r}$$



$r = \sqrt{x^2+a^2}$ is the same for all segments of the ring, so

$$U = -\frac{Gm}{r}\int dM = -\frac{GmM}{r} = -\frac{GmM}{\sqrt{x^2+a^2}}$$

(b) When $x \gg a$, $\sqrt{x^2+a^2} \to \sqrt{x^2} = x$, $U \to -\frac{GmM}{x}$

(c) $F_x = -\frac{\partial U}{\partial x} = -\frac{\partial}{\partial x}\left(-\frac{GmM}{\sqrt{x^2+a^2}}\right) = -\frac{GmMx}{(x^2+a^2)^{3/2}}$

The minus sign means the force is attractive

(d) For $x \gg a$, $(x^2+a^2)^{3/2} \to x^3$, $F_x \to -\frac{GmM}{x^2}$

(e) For $x=0$, $U = -\frac{GMm}{a}$. Each small segment of the ring is the same distance from the center and the potential is the same as that due to a point charge of mass M located at a distance a.

For $x=0$, $F_x = 0$. When the particle is at the center of the ring, symmetrically placed segments of the ring exert equal and opposite forces and the total force exerted by the ring is zero.

12.49 (a) From symmetry, the net force will be in the direction 45° from the x-axis

$$F = (6.673\times10^{-11}\ N\cdot m^2/kg^2)(0.0150\ kg)\left[\frac{2.0\ kg}{2(0.50\ m)^2} + 2\frac{1.0\ kg}{(0.50\ m)^2}\sin 45^\circ\right]$$
$$= 9.67\times10^{-12}\ N$$

(b) The initial displacement is so large that the initial potential may be taken to be zero.

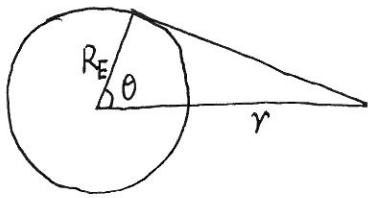
$$\frac{1}{2}mv^2 = Gm\left[\frac{2.0\ kg}{\sqrt{2}\,(0.50\ m)} + 2\frac{1.0\ kg}{0.50\ m}\right] \qquad v = 3.02\times10^{-5}\ m/s$$

12.55 (a) $T = \frac{2\pi r^{3/2}}{\sqrt{GM_E}}$, the orbital period of the earth $T = 8.64\times10^4\ s$

$$r = \left(\frac{T^2 G M_E}{4\pi^2}\right)^{\frac{1}{3}} = \left(\frac{(8.64\times10^4 s)^2(6.673\times10^{-11} N\cdot m^2/kg^2)(5.97\times10^{24}\ kg)}{4\pi^2}\right)^{\frac{1}{3}}$$
$$= 4.23\times10^7\ m$$

This is the radius of the orbit, then the height h above the earth's surface $h = r - R_E = 4.23\times10^7\ m - 6.38\times10^6\ m = 3.59\times10^7\ m$

(b)



$$\cos\theta = \frac{R_E}{r} = \frac{6.38\times10^6\,m}{4.23\times10^7\,m}$$

$$\theta = 81.3^\circ$$

The sketch shows that points at higher latitudes are blocked by the earth from viewing the satellite.

12.61 The fractional error is $1 - \dfrac{mgh}{Gmm_E\,(1/R_E - 1/(R_E+h))} = 1 - \dfrac{g}{Gm_E}(R_E+h)R_E$

Using Eq. (12.4) for g, $g = \dfrac{Gm_E}{R_E^2}$

$$1 - \frac{R_E+h}{R_E} = -h/R_E = -1\%$$

$$h = (0.01)\,R_E = 6.4\times10^4\,m$$

12.70 (a) $F_g = G\dfrac{m_E m}{R_E^2} = mg = (10.0\,kg)(9.80\,m/s^2) = 98.0\,N$

(b) The mass of the inner core is $m_{inner} = \rho_{inner}\frac{4}{3}\pi(R_2^3 - R_1^3)$

$$m_{inner} = (1.3\times10^4\,kg/m^3)\,\tfrac{4}{3}\pi\,(1.2\times10^6\,m)^3 = 9.4\times10^{22}\,kg$$

The mass of the outer core

$$m_{outer} = (1.1\times10^4\,kg/m^3)\,\tfrac{4}{3}\pi\left[(3.6\times10^6\,m)^3 - (1.2\times10^6\,m)^3\right] = 2.1\times10^{24}\,kg$$

Only the inner and outer cores contribute to the force

$$F_g = (6.67\times10^{-11}\,N\cdot m^2/kg^2)\,\frac{(9.4\times10^{22}\,kg + 2.1\times10^{24}\,kg)(10.0\,kg)}{(3.6\times10^6\,m)^2} = 110\,N$$

(c) Only the inner core contributes to the force

$$F_g = (6.67\times10^{-11}\,N\cdot m^2/kg^2)\,\frac{(9.4\times10^{22}\,kg)(10.0\,kg)}{(1.2\times10^6\,m)^2} = 44\,N$$

(d) At $r=0$, $F_g = 0$

12.72 (a) In moving to a lower orbit by whatever means, gravity does positive work, and so the speed increases.

(b) $v = (GM_E)^{\frac{1}{2}}r^{-\frac{1}{2}}$, so $\Delta v = (GM_E)^{\frac{1}{2}}\left(-\frac{-\Delta r}{2}\right)r^{-\frac{3}{2}} = \frac{\Delta r}{2}\sqrt{\frac{GM_E}{r^3}}$

Similarly, the kinetic energy is $K = \frac{1}{2}mv^2 = \frac{1}{2}G\frac{m_E m}{r}$

so $\Delta K = \frac{1}{2}\left(\frac{Gm_E m}{r^2}\right)\Delta r$ and $\Delta U = -\left(\frac{Gm_E m}{r^2}\right)\Delta r$

$$W = \Delta U + \Delta K = -\left(\frac{Gm_E m}{2r^2}\right)\Delta r$$

(c) $V = \sqrt{G m_E / r} = 7.72 \times 10^3 \text{ m/s}$, $\Delta V = \frac{\Delta r}{2}\sqrt{\frac{G m_E}{r^3}} = 28.9 \text{ m/s}$,

$E = -\frac{G m_E m}{2r} = -8.95 \times 10^{10} \text{ J}$

$\Delta K = \left(\frac{G m_E m}{2r^2}\right)\Delta r = 6.70 \times 10^8 \text{ J}$, $\Delta U = -2\Delta K = -1.34 \times 10^9 \text{ J}$

$W = -\Delta K = -6.70 \times 10^8 \text{ J}$

(d) As the term "burns up" suggests, the energy is converted to heat or is dissipated in the collisions of the debris with the ground.

12.89 The magnitude of the force that this small ring exerts on the mass m is

$(G m dM)\left(\frac{x}{(r^2+x^2)^{3/2}}\right)$

$$dF = \frac{2GmMx}{a^2}\frac{r\,dr}{(x^2+r^2)^{3/2}}$$

The total force $F = \int dF = \frac{2GmMx}{a^2}\int_0^a \frac{r\,dr}{(x^2+r^2)^{3/2}}$

$$\int_0^a \frac{r}{(x^2+r^2)^{3/2}}\,dr = \left[\frac{1}{x} - \frac{1}{\sqrt{a^2+x^2}}\right]$$

so $F = \frac{2GMm}{a^2}\left[1 - \frac{x}{\sqrt{a^2+x^2}}\right]$. The force on m is directed toward the center of the ring.

When $x \gg a$, $\frac{x}{\sqrt{a^2+x^2}} = \frac{1}{\sqrt{1+(a/x)^2}} = \left(1+\left(\frac{a}{x}\right)^2\right)^{-\frac{1}{2}} \approx 1 - \frac{1}{2}\left(\frac{a}{x}\right)^2$

$$F \approx \frac{GMm}{x^2}$$